

Mahmod Beshar Alrez

SOFTWARE ENGINEER

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Professional Summary

Versatile software engineer with over 5 years of experience in full-stack web and mobile application development. Skilled in delivering scalable, user-focused solutions by translating complex requirements into robust technical architectures. Strong communicator with a proven track record of collaborating across cross-functional teams. Passionate about emerging technologies and continuous improvement.

Experience

Elizza & CO | Software Engineer

January 2022 – Feb 2025

- Developed a CRM system tailored for jewelry businesses, including modules for products, contracts, accounting, and customer management.
- Gathered requirements and led sprint planning to align development with business goals.
- Led the reengineering of the CRM front end by migrating from Blade templates to Vue 3, resulting in a 40% boost in performance and maintainability.
- Enhanced system functionality and UX through full-stack improvements.
- Authored comprehensive technical documentation to streamline onboarding for new developers.

BigLedger Sdn Bhd | Software Engineer

December 2019 – September 2021

- Built a cross-platform CRM mobile app for a major automotive client, ensuring seamless use on both Android and iOS.
- Developed and maintained asynchronous RESTful APIs using Spring Boot and Java 8.
- Improved content management efficiency by enhancing CMS modules using Angular 8, HTML, and CSS.
- Upgraded mobile app UI using Ionic 3, significantly enhancing user experience.
- Integrated Angular components as micro frontends to support modular web architecture.
- Reduced support issue resolution time by streamlining maintenance workflows.
- Trained junior developers to expedite onboarding and enhance team productivity.
- Maintained clear and comprehensive development documentation.

Inai Kiara Sdn Bhd | Software Engineer

September 2018 – December 2019

- Developed a custom web-based system to automate internal workflows, greatly improving operational efficiency.
- Participated in Agile-based development, from requirement gathering to system deployment.
- Built custom dashboards using Laravel (backend) and Vue.js (frontend).
- Achieved an 80% reduction in processing time across key departments, including Procurement, Inventory, Trips, Crewing, and Task Management.
- Documented system architecture and deployment processes for future developers.

Education

Master of Science in Computer Science

Central Michigan University, USA

Graduated: 2025

GPA: 3.6 / 4.0

Bachelor of Computer Science in Software Engineering

Multimedia University, Malaysia

Graduated: 2018

GPA: 3.25 / 4.0

Skills & Technologies

- Programming Languages: Java, PHP, JavaScript, TypeScript, Python
- Frameworks & Libraries: Laravel, Spring Boot, FastAPI, Angular, Vue.js, Ionic,
- DevOps & Tools: Git, Jenkins, Docker, CI/CD Pipelines, AWS
- Databases: MySQL, PostgreSQL
- Front-End: HTML, CSS, JavaScript, SASS, Responsive Design
- Cloud Platforms: AWS, Azure

Projects

AI-powered recipe suggestion system.

Developed an AI-powered recipe suggestion system using FastAPI and a custom-trained language model, allowing users to input ingredients and receive personalized meal ideas. Integrated Vue.js on the frontend and optimized API response times through model hosting on Google Colab.

E-commerce Platform for Underwear Brand.

- Designed and developed a fully functional e-commerce platform using Laravel and Vue.js.
- Improved client's digital presence and sales operations, resulting in positive customer feedback and sales growth.

ERP System for Textile Manufacturers.

- Developed an ERP solution using Laravel and Vue.js to address production and inventory challenges.
- Successfully deployed for multiple manufacturers, enhancing their operational efficiency.

Online Restaurant Ordering System.

- Built a contactless ordering system during the COVID-19 pandemic using Laravel and Vue.js.
- Enabled restaurants to manage orders digitally and safely, reducing direct contact between staff and customers.

Cover Letter

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10/25/2025

Hiring Committee

University of Michigan – College of Engineering, Robotics Department

Ann Arbor, MI

Dear Hiring Committee,

I am writing to express my interest in the Software Engineer position within the Robotics Department at the University of Michigan. I recently earned my Master of Science in Computer Science from Central Michigan University (May 2025) and bring over five years of professional experience developing and maintaining full-stack applications. My technical background, coupled with a passion for innovation and academic collaboration, aligns strongly with the Robotics Department's mission to advance research and technology for the public good.

In my role at Elizza & Co, I led the design and development of a custom CRM platform, boosting system performance by 40% through optimized backend architecture and modular front-end refactoring using Laravel, Vue.js, and MySQL. I also managed deployments on AWS, authored system documentation, and collaborated closely with non-technical stakeholders to deliver reliable, user-friendly software solutions. Earlier, at BigLedger Sdn Bhd, I built and maintained RESTful APIs using Java Spring Boot, improving data synchronization and backend efficiency across large-scale enterprise systems.

Beyond traditional software engineering, I have also developed an AI-driven Recipe Suggestion System, integrating a fine-tuned language model with a FastAPI backend and structured SQL data. This project deepened my understanding of AI integration, model optimization, and data pipeline design, skills that directly support the Robotics Department's data-intensive, research-driven initiatives.

I am drawn to Michigan Engineering's commitment to excellence, inclusion, and impact. I'm eager to contribute my technical expertise in Python, Java, AWS, and modern web frameworks to help build secure, scalable applications that enhance research productivity and the student experience.

Thank you for considering my application. I would welcome the opportunity to contribute my skills, problem-solving mindset, and dedication to innovation to the Robotics Department's mission.

Sincerely,

Mahmod Beshar Alrez